

ME 462 Final Project

Predictive Torque Control with Battery Aware constraints for Electric Bicycle Safety and Longevity



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December 12, 2025

I. Introduction

Diagnosing and troubleshooting e-bike motor controllers has created significant costs and operational delays for company Green Riders. Green Riders operates a fleet of ~300 bikes with informal workers who ride around the city of Cape Town to deliver Uber Eats and other food aggregators. Faulty motor controllers were frequently caused by blown capacitors, which could only be diagnosed through destructive measuring. This process was lengthy and ended with the controller being discarded. In this project, a preventative supervisory control strategy is developed to mitigate capacitor overloading through constrained torque and current regulation. Human input, environmental disturbances, electrical constraints, and fault tolerance are necessary to incorporate into the control methodology to create a safe operating experience. The primary motivation of this project is to design a control system that constrains the bicycle to a maximum allowable speed while ensuring smooth torque and current ramp-up during startup, controlled deceleration and stopping behavior, reduced battery stress through current shaping, and robustness to disturbances such as hills and environmental factors such as different terrain, and load requirements. Based on experience, the voltage measurement of the bike cannot be measured online but rather uploaded to the controller periodically via an IoT process. The system will primarily utilize online battery current and speed to control the torque of the motor. The manipulated variables will thus be the PWM duty command, which indirectly regulates motor current and torque. The output of the system will be speed of the e-bike. The control framework will predict load changes and regulate motor behavior.

The methodology used in this project is largely a system level approach with empirical methods of control. While more complicated control methods can be utilized to achieve desired results, short term improvements were suggested to solve a persistent issue with controller damage. In the long term, this method could plateau.

II. Model and Control Overview

Assume, strictly 1 dimensional travel on the bike. Any uphill or downhill changes in road grade will be considered F_{load} on the bike. The longitudinal dynamics of the bicycle are

modeled using Newton's second law, with all external resistive effects lumped into a load force term:

$$m\dot{v} = \frac{\tau_m}{r_w} - F_{load}. \quad (1)$$

The hill disturbance can be modeled with

$$F_{hill} = mgsin(\theta). \quad (2)$$

This term is unknown a priori and is lumped into the total load force together with aerodynamic drag and rolling resistance.

The motor that is used on the e-bikes is the BafangG062 with the following specifications [1]:

Table 1

Wheel Diameter (Inch)	20 / 26
Nominal Voltage (Vdc)	48, 48 / 52, 48 / 52
Rated Power (W)	250, 500, 750
Speed Limit (km/h)	25 / 32, 25-35, 45
Max. Possible Torque (Nm)	55, 65, 80
Color	Black
Weight (kg)	4.2, 4.5, 4.9

Additional model specifications are outlined in table 2.

Table 2

Max Speed	V_{max}	45km/h = 12.5 m/s
Wheel radius	r_w	0.6604 m
Wheel angular speed at limit	ω_{max}	18.93 rad/s ($\approx$ 181 rpm
Rated Power	P_{nom}	750 W
Grade Range	θ_{grade}	0-15%
Total Mass	m	80+m _r kg where m _r is the rider mass

The system is modeled with two states, $x = [\omega, \tau_m]$, ω is the wheel/motor angular speed [rad/s] and τ_m is the motor torque delivered at the hub [N·m]. The input $u = d \in [0,1]$, where d is the PWM duty command allowing the control methodology to use square waves as in controls theory instruction. The battery current,

$$i_b \approx \frac{\tau_m \omega}{\eta V_b}. \quad (3)$$

The PWM duty cycle represents the fraction of time the battery voltage is applied to the motor terminals. At the supervisory timescale, the motor responds to the average applied voltage, given by $V_{\text{motor}} = dV_{\text{bat}}$. The resulting motor current is determined by the balance between this applied voltage, the speed-dependent back-EMF $K_e \omega$ and the motor resistance, yielding $I = \frac{dV_{\text{bat}} - K_e \omega}{R}$. Motor torque is proportional to current via $\tau = K_t I$, providing a direct physical link between PWM duty and mechanical output. This relationship explains the strong coupling between duty, current, and speed observed in simulation and motivates the inclusion of explicit current-limiting mechanisms. speed, $v = r_w \omega$. While η is the lumped efficiency and V_b is held constant between voltage samples. The dynamics are further defined as follows,

$$\dot{\omega} = \frac{1}{J_{eq}} (\tau_m - \tau_{load}(\omega, \theta, m)). \quad (4)$$

The torque command from duty is $\tau_{cmd}(d) = \text{sat}_{[0, \tau_{max}]}(\tau_{max} d)$. The actuator dynamics is

$$\dot{\tau}_m = \frac{1}{T_T} (\tau_{cmd}(d) - \tau_m). \quad (5)$$

A block level diagram is detailed in figure 1 with an ensuing explanation.

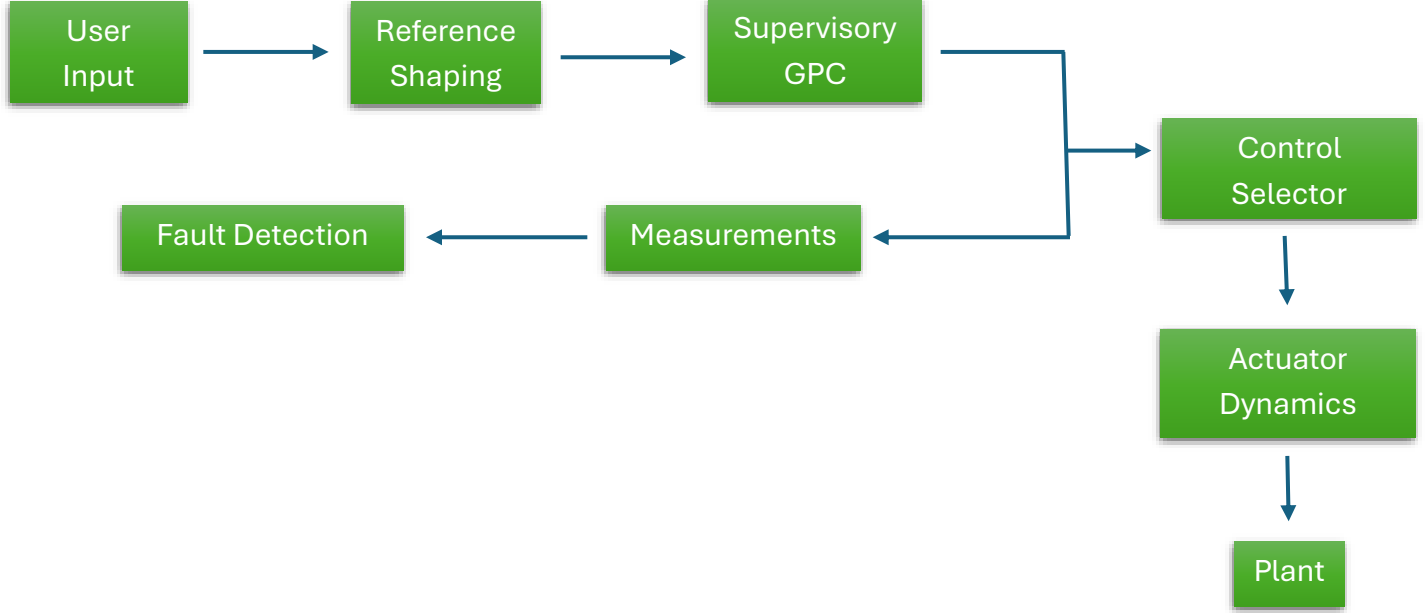


Figure 1

The supervisory controller operates at a slow sampling rate and modifies the PWM duty command sent to the motor controller, which remains responsible for fast inner-loop current and voltage regulation. Rather, it provides an accessory to increase the operational efficiency of certain e-bikes. Some riders may “upgrade” their bike with this controller for better throttle, efficiency, and battery life with this supervisory control system. This ensures robustness to sensor faults, smooth and safe actuation independent of optimization success, and fault detection and mode switching.

The prediction horizon dynamics is

$$\hat{\mathbf{x}}_{k+i+1} = \mathbf{f}(\hat{\mathbf{x}}_{k+i}, \mathbf{d}_{k+1}) \quad i = 0, \dots, N_p - 1. \quad (6)$$

The predicted speed output:

$$\hat{v}_{k+i} = \mathbf{r}_w \hat{\boldsymbol{\omega}}_{k+1}. \quad (7)$$

The constraints are outlined in table 3.

Table 3

Speed Constraint	$\hat{v}_{k+1} \leq v_{max}$
Torque Constraint	$0 \leq \hat{t}_{m,k+i} \leq 80 \text{ Nm}$

Battery Current Constraint	$\hat{i}_{b,k+1} = \frac{\hat{\tau}_{m,k+i}\hat{\omega}_{k+i}}{\eta V_b} \leq \begin{pmatrix} 23 A & \text{continuous} \\ 30 A & \text{burst} \end{pmatrix}$
Duty Slew Constraint	$ \Delta_{k+i} \leq \Delta d_{max}$

Fault detection and identification are implemented using a residual based consistency check outlined in Fault diagnosis and accommodation... [2]. These are shown in table 4:

Table 4

Torque-current residual	$r_\tau(k) = \tau_{meas}(k) - \tau_m(k)$
Speed Prediction residual	$r_\omega(k) = \omega_{meas}(k) - \hat{\omega}_{k k-1}$

This approach avoids online parameter identification and instead exploits physical inconsistency between measured current, torque, and speed along with predictable rider behavior. Two modes will be used for fault-tolerant control and switching logic. Mode 1 is normal GPC active predictive control that enforces speed limit and smooth power draw. Mode 2 is the fallback user-controlled bypassed state.

This control method allows tolerance for noisy current sensing especially considering thermal drift. The controls will be compatible with embedded hardware and the internal clock will not be power or calculation intensive. Optimization is safe under failure and the system can switch to user inputs. The system is scalable with room for improvement and data collection for analytics in fleet efficiency.

III. Theoretical Results

The dominant disturbance is road grade modeled as a bounded slope: $g_r \in [0, 0.15]$, $\theta = \arctan(g_r)$. This means that the grade force is bounded

$$\begin{aligned} 0 \leq F_{hill} &= mgsin\theta \\ &\leq mgsin(\arctan(0.15)) \end{aligned} \tag{8}$$

and thus the load torque is bounded:

$$0 \leq \tau_{hill} = r_\omega F_{hill} \leq r_\omega mgsin(\arctan(0.15)) \tag{9}$$

Constraint-based controllers are particularly effective when disturbances are bounded, as these bounds correspond to physically meaningful operating limits..

Rider mass is treated as an uncertain but constant parameter, representing structured parametric uncertainty. The controller can be validated against worst case combinations while testing for performance at average or nominal values.

The principal safety requirement is a hard speed cap of 45 km/h. The actuator constraints are based on the battery specifications. The optimization must be feasible for the GPC to take effect. Safety is embedded as an inequality, not something that must be tuned allowing for uses across geographies.

The road grade and mass are bounded, thus the worst-case load torque occurs at the maximum torque, maximum mass, and maximum speed. The feasibility of this scenario must lie in a range where the torque is less than the maximum capacity of the motor. Regardless, the max speed will not be exceeded.

Speed measurement noise is modeled as a gaussian measurement noise

$$\mathbf{n}_n \sim \mathcal{N}(\mathbf{0}, \sigma_v^2) \quad (10)$$

The resolution of this measurement is 0.1 km/h after viewing it on the tachometer. Implicit disturbance even without slope estimation, GPC reacts to persistent error caused by τ_{load} by adjusting duty. Under slowly varying grade change, receding horizon control provides disturbance rejection through repeated re-optimization.

Fault tolerant switching rule is implemented if current measurement violates limits or becomes inconsistent with torque/speed behavior. There is a persistence requirement of this where the fault must hold for N steps which can be tuned for safety considerations based on model behavior at failure. I will introduce a dwell time as well in which there must be a minimum time in the fallback mode before switching back to MPC control with the correct threshold for entry.

To summarize, the system may not be able to hold the speed requirement at maximum load. Voltage cannot be used for real time power estimation; therefore current related constraints rely primarily on current sensing and margins. A large time step $T_s=5$ s introduces a supervisory role

of the controller. Fast, transient shaping of current is delegated to motor controller. This control system will act as a safety mechanism to prevent the controller from being damaged.

IV. Preliminary Simulation Results

Firstly, the model must reflect real world deceleration effects, otherwise creating an unbounded velocity response. Thus a simple aerodynamic and rolling resistance is introduced in the model along with some simplified motor dynamics.

$$F_{roll} = C_r m_{total} g \sin(\theta) \quad (11)$$

The rolling resistance coefficient is taken to be some nominal value, 0.015. In figure 2, preliminary simulation results for this control methodology is displayed.

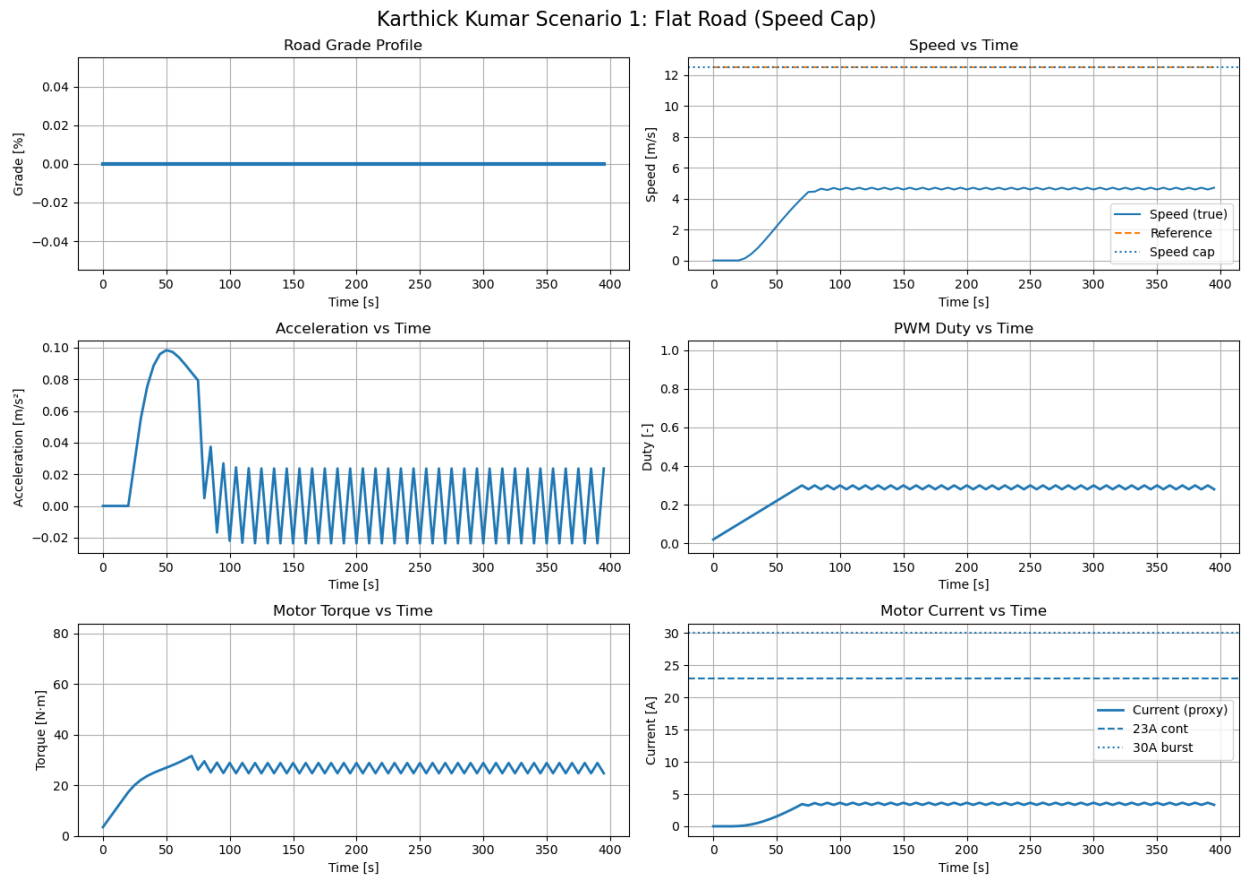


Figure 2 Preliminary Results for Flat Road

In the example, a flat road is used to simulate speed vs time as well as acceleration vs time. The results here look promising, but time is valuable and in a straight line, riders will be expecting to go as fast as possible. When tuning the model by decreasing lambda, a less ergonomic and somewhat dangerous output arises in figure 3:

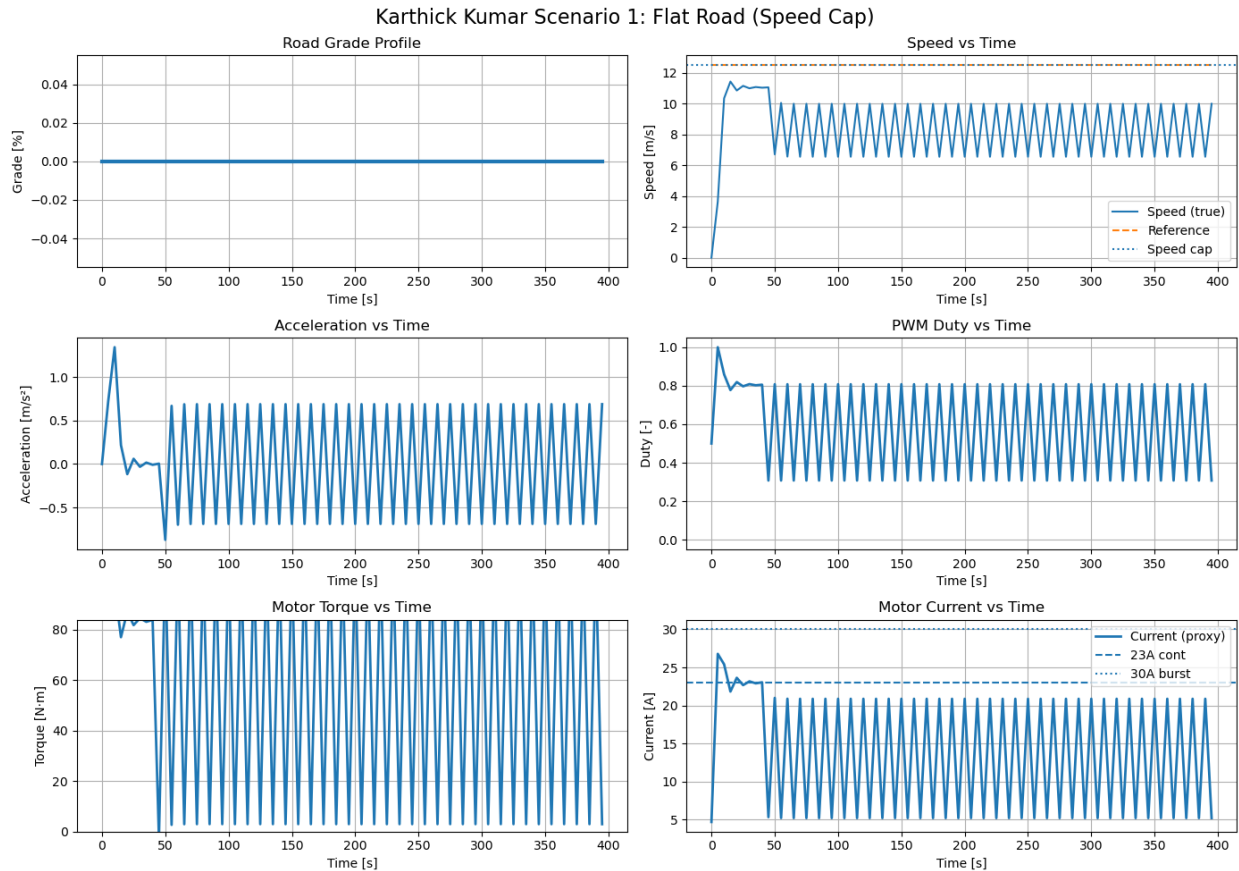


Figure 3 $Nu = 4$, $\lambda = 0.5$, Flat Road

In the figure, the current behavior is within limits. Torque behavior is operating just above the motor rated limit and the but the acceleration is dangerous for the rider. This is most likely an issue with the sampling time because the deceleration acts at every time step according to current logic. When decreasing T_s from 5s to 1s, figure 4 is generated.

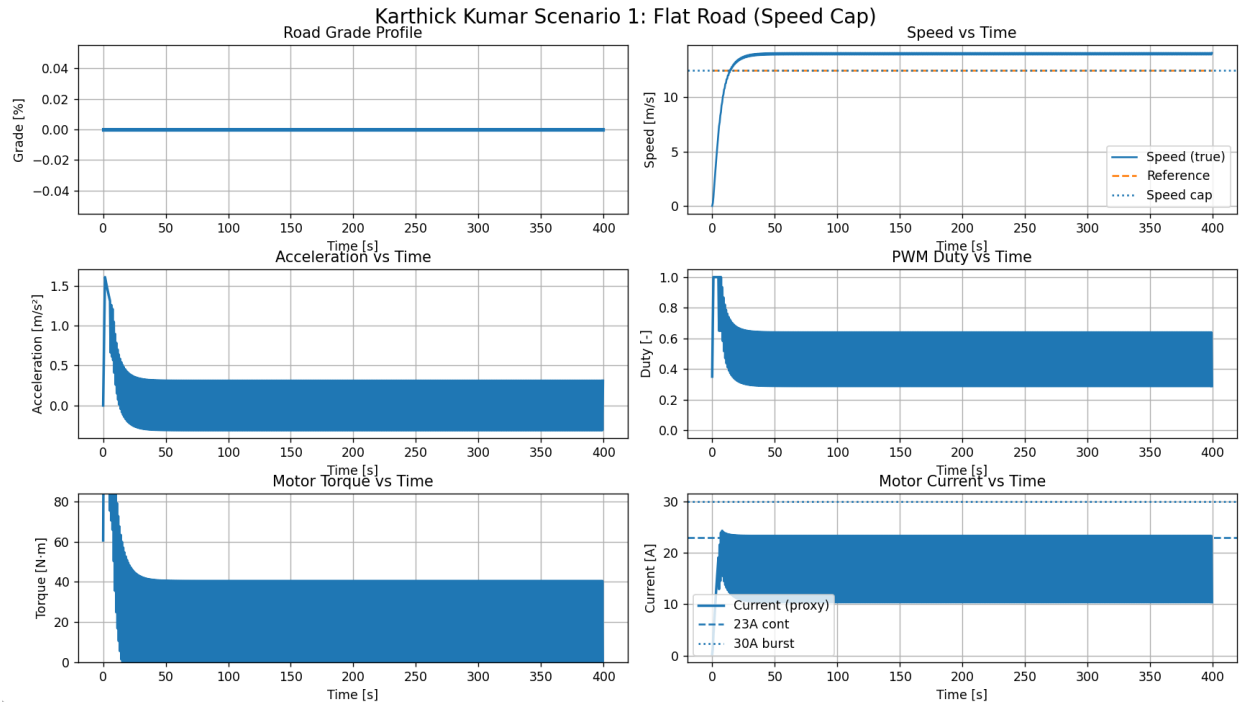


Figure 4 Low T_s , $Nu = 4$, $\lambda = 0.5$

This system is surpassing the speed reference which is cause of concern; however, the current does not pass the continuous current threshold. The lower time step has now made these charts much harder to read, but surprisingly creates a visually smooth speed

chart. These results are very promising if the clock and controller are efficient enough to run at this speed without creating significant power loss.

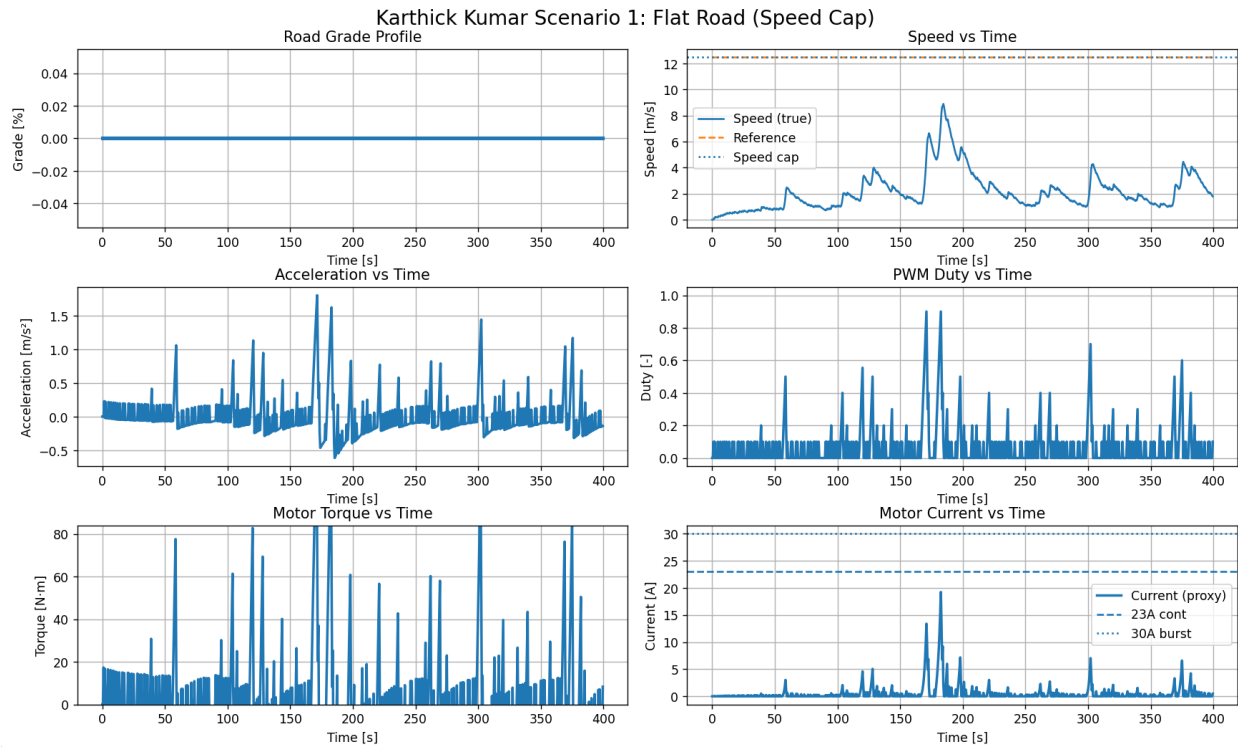


Figure 5 $Nu = 50$ $\lambda = 0.01$

Surprising results are seen in figure 5 but upon inspection, the hi and lo for the PWM duty must be increased to allow for more range of control. In figure 6, a quite successful control output is generated with the same parameters as figure 5 but increased constraints on lo and hi which bound $\Delta duty$ per sample.

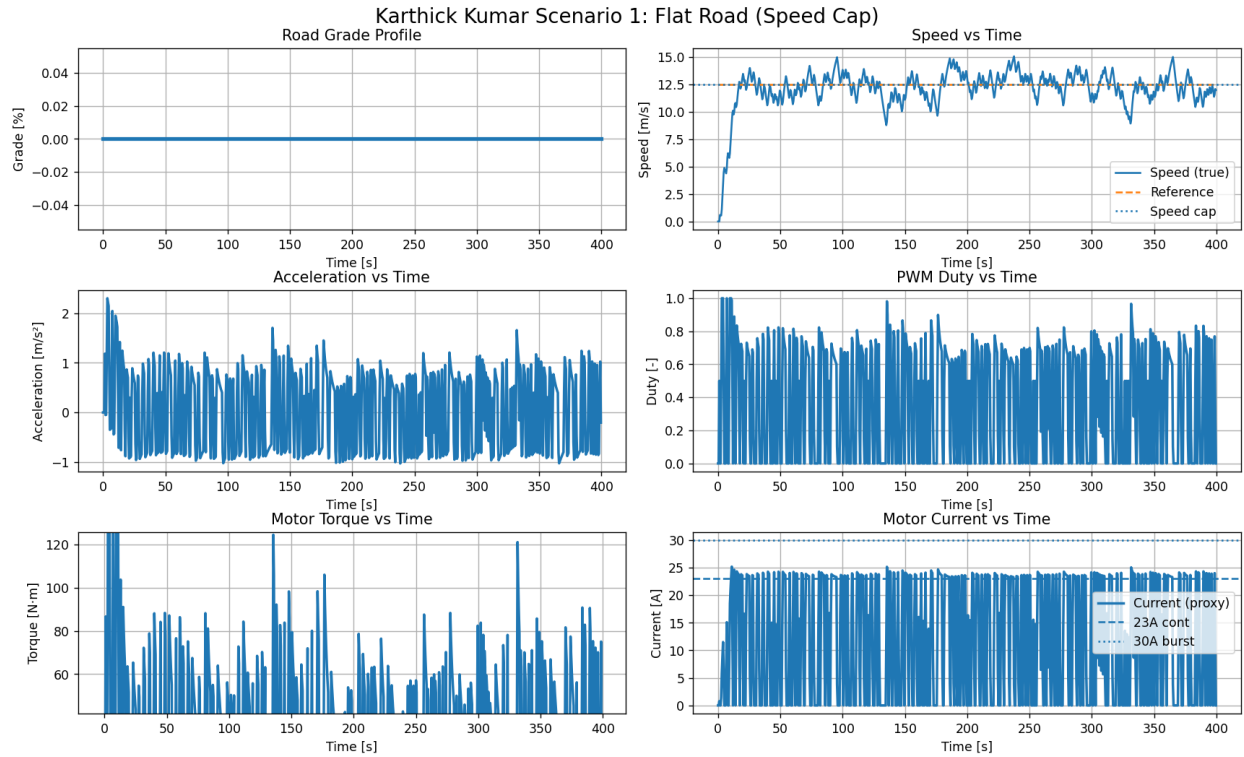


Figure 6 Successful flat road control

The current stays within range of the reference and allows to rider to speedily go along the flat road. Moreover, the current is within acceptable range. While the torque is over the rated 80 Nm, this may simply be a product of arbitrary rolling resistance coefficient. The duty cycle is similar to what a real world duty cycle may look like. This control system should be wired before the motor controller to prevent damage to the controller from bad signals from direct input and current surge.

In figure 7, another scenario is explored where the bike approaches a steep hill. In this control, the bike is unable to go up the hill at maximum grade.

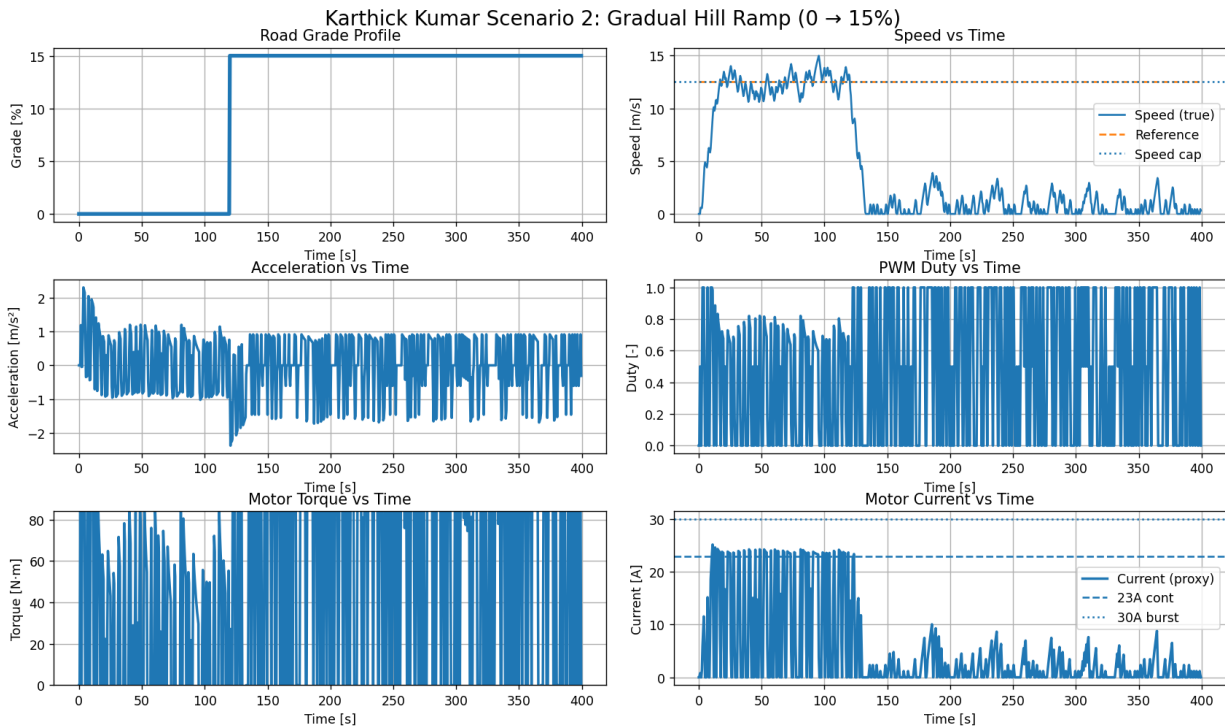


Figure 7 Scenario two, increasing road grade

However, when decreasing the grade, a more agreeable control signal is given seen in figure 8.

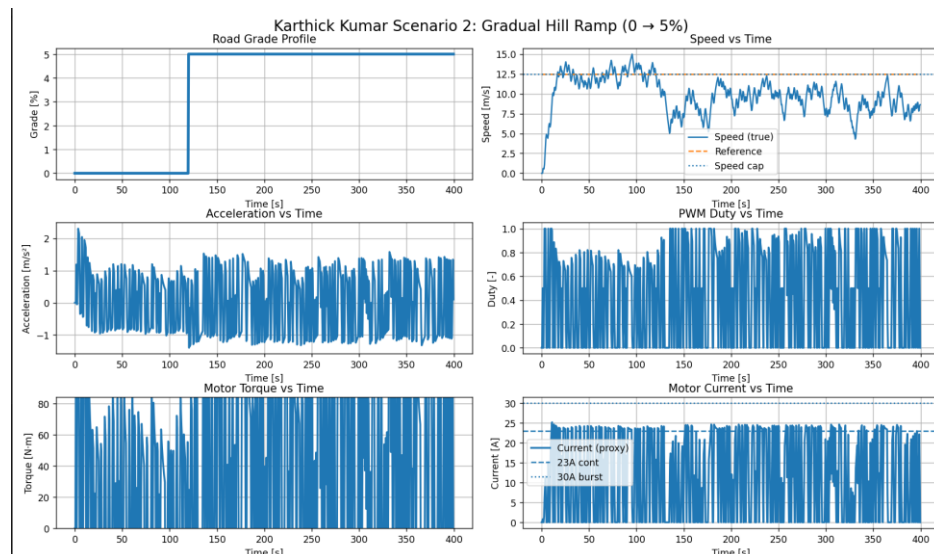


Figure 8 Scenario 2 Successful control

Automatically decreasing the speed on the hill is a great way to ensure safety when attacking the hill. Again, the current requirement is reached.

In this situation, the fault tolerant mechanism should kick in. Once the power idles at 0 for 1 second, the rider will see the road grade and thus have a user input relative to the road grade. The next section will implement this.

Fault Tolerance:

When the system enters a power-limited oscillatory stall, the supervisory controller may incorrectly reduce duty despite available current headroom. A temporary manual override increases applied voltage to exploit short-term torque capability, allowing the vehicle to escape the stalled operating point before resuming predictive control.

For instance, in the case of a motor stall, the controller can send a burst current to the motor controller, protecting the electronics while allowing the bike to create the necessary acceleration.

Fault tolerant control of figure 7 is displayed in figure 9.

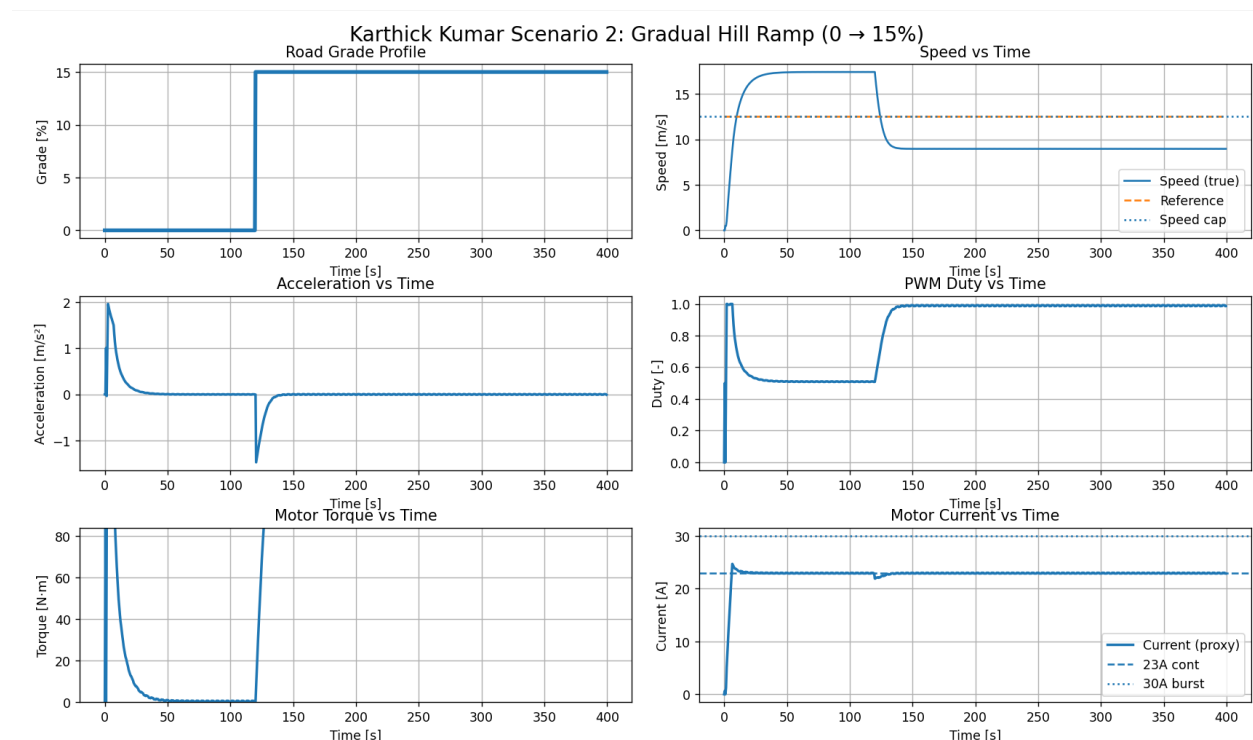


Figure 9

The fault tolerance activates when steep grades or control failures cause large residuals, switching to a maximum current available mode to provide maximum available power.

Similarly, a fault tolerance model of figure 6 is implemented in figure 10:

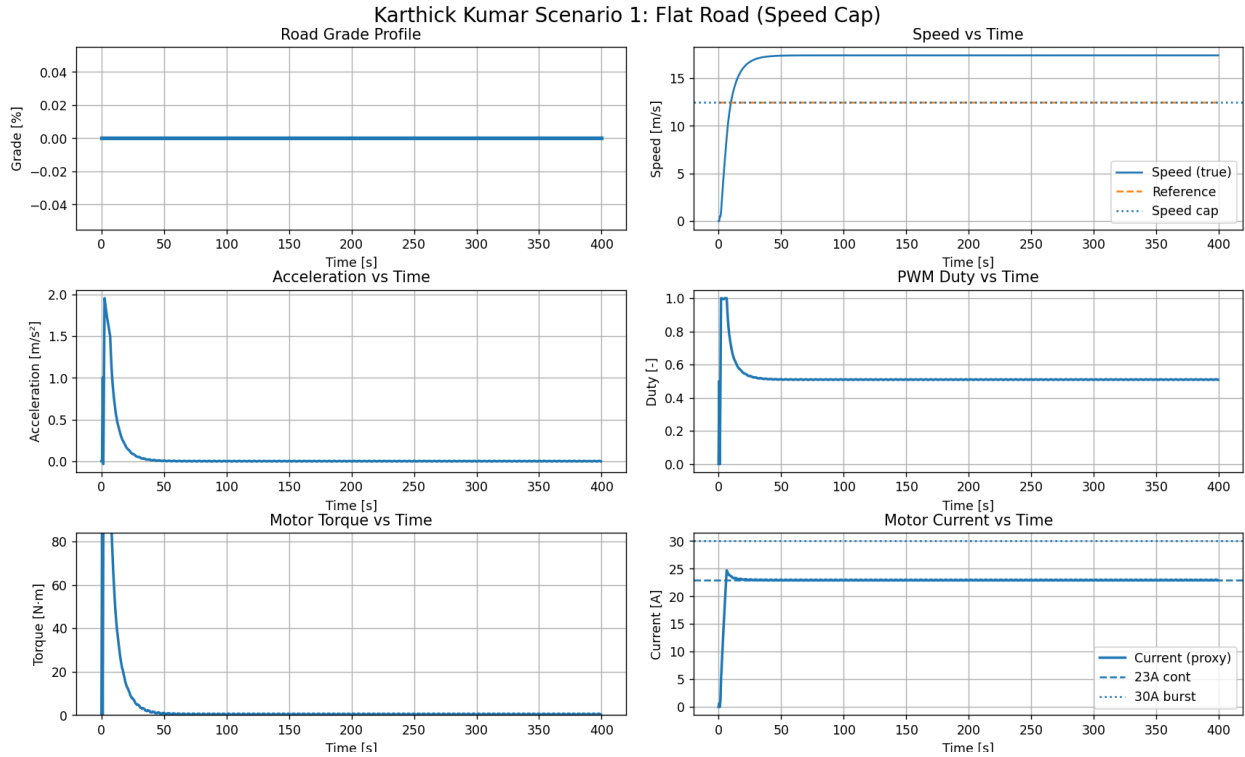


Figure 10

The PWM duty cycle is effectively maintained to keep the current at the maximum once it recognizes the need to turn off GPC. Further testing, implementing, and more stringent entry requirements are needed to make this fault tolerant mode more practical.

Ideally, in figure 7, the fault tolerant control would recognize the gap in max current and current, compared to reference speed and speed and would apply a ramp up current at around time 150s to avoid the motor stall and GPC failure. In many cases in lesser loads i.e. lesser road grade, the GPC works fine.

V. Conclusion

This project demonstrated the feasibility of a supervisory predictive control framework for electric bicycle safety and battery protection using Generalized Predictive Control (GPC). By explicitly modeling rolling resistance, road grade, and current limitations, the controller successfully constrained vehicle speed and regulated power demand under nominal operating conditions. The use of PWM duty as the manipulated variable allowed the controller to remain non-intrusive to the motor controller's inner loops while still enforcing high-level safety and longevity objectives.

A key contribution of this work is the identification and handling of power-limited and stalled operating regimes. When the predictive controller encounters conditions where no feasible control action can improve speed—such as steep grades or heavy loading—the system tries to transition to a fault-tolerant manual override mode. This mode applies maximum safe current to guarantee recoverability from stalled states while respecting electrical limits. The separation between nominal predictive control and emergency high-effort control prevented oscillatory behavior and ensured robust performance across a wide range of operating scenarios.

Overall, the results show that supervisory control can significantly improve operational safety and diagnostic capability in low-cost electric vehicle platforms without requiring high-bandwidth sensing, invasive hardware changes, or online system identification.